HSC EXE Packaging Guide

Step-by-step guide to build and package the Windows installer EXE for the desktop GIS app.

IT Whitelist URLs (before setup)

Ask client IT team to allow outbound access to these domains before running `yarn install` and `yarn package` :

```
https://registry.npmjs.org
https://registry.yarnpkg.com
https://yarnpkg.com
https://classic.yarnpkg.com
https://github.com
https://api.github.com
https://objects.githubusercontent.com
https://github-releases.githubusercontent.com
https://release-assets.githubusercontent.com
https://electronjs.org
https://www.electronjs.org
https://github.com/electron/electron/releases
https://github.com/electron-userland/electron-builder-binaries/releases
```

> **Why needed:** package install/downloads use npm/yarn registries, and Electron/electron-builder binaries are commonly downloaded from GitHub release/CDN endpoints.

> **Optional (only if client also performs Android build/sync steps):** whitelist these repositories too: `https://maven.google.com/` , `https://repo1.maven.org/maven2/` , `https://services.gradle.org/` .

Build Environment Prerequisites

- **Node.js:** `22.12.0`
- **Yarn:** `1.22.22`
- **Other package versions:** from `package.json`

```
# Verify versions
node -v
yarn -v

# Install yarn globally if missing
npm install -g yarn
```

> **TLS/HTTPS requirement:** secure HTTPS traffic must be allowed by client IT/network policy, or dependency/package downloads can fail.

> **Root naming requirement:** In this document, root is referred to as `hsc-android` (as requested), even though desktop packaging commands run from the desktop project folder.

Project Structure Assumed

```
hsc-android/                          <- shared package root sent to client
├── package.json
├── electron/
├── src/
└── ... (rest of desktop project files)
```

> **Important:** Client receives only this desktop project folder, renamed as `hsc-android` .

Where to place offline tiles

The desktop app auto-reads tiles from the current Windows user's `Documents\tiles` folder.

```
Default tile folder (Windows):
C:\Users\\Documents\tiles

Equivalent using environment variable:
%USERPROFILE%\Documents\tiles
```

Minimum expected files/folders inside `tiles` :

```
tiles/
├── style.json
├── sprites/            (if style references sprites)
├── fonts/              (if style references glyphs)
└── ... tile data referenced by style.json
```

> **Important:** `style.json` is required at the root of `Documents\tiles` . If missing, map style loading fails.

Step 1: Open PowerShell and go to the shared package folder

```
# If you are already in the parent directory:
cd "hsc-android"

# Or use full path wherever client extracted it:
cd "D:\any-folder\hsc-android"
```

Step 2: Install dependencies

```
yarn install
```

Step 3: Package EXE

```
yarn package
```

> **Client flow:** run `yarn install` first, then `yarn package` . `yarn package` handles required build steps internally.

Where output EXEs are generated

- **Installer EXE:** `release\HSC GIS Desktop Setup 1.0.0.exe`
- **Unpacked EXE:** `release\win-unpacked\HSC GIS Desktop.exe`

> **Success check:** after packaging, installer EXE should exist in `release\` and the installed app should launch without missing CSS/JS file errors.

Troubleshooting (cache fix for yarn package)

If `yarn package` fails with NSIS cache/elevate errors, run:

```
cd "hsc-android"

Get-Process node,electron,electron-builder,signtool -ErrorAction SilentlyContinue | Stop-Process -Force

Remove-Item "$env:LOCALAPPDATA\electron-builder\Cache\nsis" -Recurse -Force -ErrorAction SilentlyContinue
Remove-Item "$env:LOCALAPPDATA\electron-builder\Cache\nsis-resources" -Recurse -Force -ErrorAction Silent
Remove-Item ".\release" -Recurse -Force -ErrorAction SilentlyContinue

New-Item -ItemType Directory -Force ".\.cache\electron-builder" | Out-Null
$env:ELECTRON_BUILDER_CACHE = (Resolve-Path ".\.cache\electron-builder").Path

yarn package
```

.gitignore entries to add (client side)

Ensure these entries exist in `.gitignore` to avoid committing Windows/Electron generated files:

```
# Electron build artifacts
dist-electron/
release/

# Electron builder local cache
.cache/

# JVM crash/replay logs (can be generated by Gradle/Android tooling on Windows)
android/hs_err_pid*.log
android/replay_pid*.log
```